SYSC 5108 Exam Study
Assignment 4, Question 8

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Question

Let $X = (X_1, X_2)$ be a two-dimensional random variable with binary entries

$$X_1, X_2 \in \{0, 1\}.$$

The joint probabilities are

$$P(X_1, X_2) = \begin{bmatrix} P(0,0) \\ P(0,1) \\ P(1,0) \\ P(1,1) \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \\ \frac{1}{2} \\ \frac{1}{8} \\ \frac{1}{8} \end{bmatrix}.$$

(a) Find the distributions $P(X_1 \mid X_2)$ and $P(X_2 \mid X_1)$.

(b) Run Gibbs sampling for 3 iterations starting from uniform distributions for both X_1 and X_2 independently, using the random numbers

$$0.94,\ 0.27,\ 0.30,\ 0.18,\ 0.75,\ 0.35,\ 0.61,\ 0.42, \ldots$$

Course and Textbook Cross-References

- **Chapter 7 slides, Gibbs Sampling:** repeatedly sample one variable from its conditional distribution given the other variables.
- **Chapter 7 slides, Sampling for Structured Models:** Gibbs sampling uses conditional distributions $p(x_j \mid x_{-j})$.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 17:** Markov chain Monte Carlo updates the state by repeated sampling from conditional distributions.

Step 1: Write the Joint Table Clearly

The lexicographic ordering means

$$P(0,0) = \frac{1}{4}, \qquad P(0,1) = \frac{1}{2}, \qquad P(1,0) = \frac{1}{8}, \qquad P(1,1) = \frac{1}{8}.$$

So the joint table is

	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	$\frac{1}{4}$	$\frac{1}{2}$
$X_1 = 1$	$\frac{1}{8}$	$\frac{1}{8}$

Part (a): Find $P(X_1 \mid X_2)$ and $P(X_2 \mid X_1)$

Step 2: Compute the Marginals

For X_2,

$$P(X_2 = 0) = P(0,0) + P(1,0) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8},$$

$$P(X_2 = 1) = P(0,1) + P(1,1) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}.$$

For X_1,

$$P(X_1 = 0) = P(0,0) + P(0,1) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4},$$

$$P(X_1 = 1) = P(1,0) + P(1,1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Step 3: Compute $P(X_1 \mid X_2)$

When $X_2 = 0$,

$$P(X_1 = 0 \mid X_2 = 0) = \frac{P(0,0)}{P(X_2 = 0)} = \frac{\frac{1}{4}}{\frac{3}{8}} = \frac{2}{3},$$

$$P(X_1 = 1 \mid X_2 = 0) = \frac{P(1,0)}{P(X_2 = 0)} = \frac{\frac{1}{8}}{\frac{3}{8}} = \frac{1}{3}.$$

Hence

$$P(X_1 \mid X_2 = 0) = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}.$$

When $X_2 = 1$,

$$P(X_1 = 0 \mid X_2 = 1) = \frac{P(0,1)}{P(X_2 = 1)} = \frac{\frac{1}{2}}{\frac{5}{8}} = \frac{4}{5},$$

$$P(X_1 = 1 \mid X_2 = 1) = \frac{P(1,1)}{P(X_2 = 1)} = \frac{\frac{1}{8}}{\frac{5}{8}} = \frac{1}{5}.$$

Hence

$$P(X_1 \mid X_2 = 1) = \begin{bmatrix} \frac{4}{5} \\ \frac{1}{5} \end{bmatrix}.$$

Step 4: Compute $P(X_2 \mid X_1)$

When $X_1 = 0$,

$$P(X_2 = 0 \mid X_1 = 0) = \frac{P(0,0)}{P(X_1 = 0)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3},$$

$$P(X_2 = 1 \mid X_1 = 0) = \frac{P(0,1)}{P(X_1 = 0)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}.$$

Hence

$$P(X_2 \mid X_1 = 0) = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}.$$

When $X_1 = 1$,

$$P(X_2 = 0 \mid X_1 = 1) = \frac{P(1,0)}{P(X_1 = 1)} = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{1}{2},$$

$$P(X_2 = 1 \mid X_1 = 1) = \frac{P(1,1)}{P(X_1 = 1)} = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{1}{2}.$$

Hence

$$P(X_2 \mid X_1 = 1) = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Answer for Part (a)

$$\boxed{P(X_1 \mid X_2 = 0) = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}, \qquad P(X_1 \mid X_2 = 1) = \begin{bmatrix} \frac{4}{5} \\ \frac{1}{5} \end{bmatrix}}$$

$$\boxed{P(X_2 \mid X_1 = 0) = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}, \qquad P(X_2 \mid X_1 = 1) = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}}$$

Part (b): Gibbs Sampling for 3 Iterations

Step 5: Initialization from Independent Uniform Distributions

The question says to start with uniform distributions for X_1 and X_2 independently. For a binary variable, that means

$$P(X_j = 0) = P(X_j = 1) = \frac{1}{2}.$$

Using the first two random numbers:

- $r_1 = 0.94$. Since $0.94 > 0.5$, take
$$X_1^{(0)} = 1.$$
- $r_2 = 0.27$. Since $0.27 \leq 0.5$, take
$$X_2^{(0)} = 0.$$

So the initial state is

$$(X_1^{(0)}, X_2^{(0)}) = (1, 0).$$

Step 6: Iteration 1

Sample $X_1^{(1)}$ from $P(X_1 \mid X_2 = 0)$:

$$P(X_1 = 0 \mid X_2 = 0) = \frac{2}{3}.$$

Use $r_3 = 0.30$. Since $0.30 \leq \frac{2}{3}$,

$$X_1^{(1)} = 0.$$

Now sample $X_2^{(1)}$ from $P(X_2 \mid X_1 = 0)$:

$$P(X_2 = 0 \mid X_1 = 0) = \frac{1}{3}.$$

Use $r_4 = 0.18$. Since $0.18 \leq \frac{1}{3}$,

$$X_2^{(1)} = 0.$$

So

$$(X_1^{(1)}, X_2^{(1)}) = (0, 0).$$

Step 7: Iteration 2

Sample $X_1^{(2)}$ from $P(X_1 \mid X_2 = 0)$:

$$P(X_1 = 0 \mid X_2 = 0) = \frac{2}{3}.$$

Use $r_5 = 0.75$. Since $0.75 > \frac{2}{3}$,

$$X_1^{(2)} = 1.$$

Now sample $X_2^{(2)}$ from $P(X_2 \mid X_1 = 1)$:

$$P(X_2 = 0 \mid X_1 = 1) = \frac{1}{2}.$$

Use $r_6 = 0.35$. Since $0.35 \leq \frac{1}{2}$,

$$X_2^{(2)} = 0.$$

So

$$(X_1^{(2)}, X_2^{(2)}) = (1, 0).$$

Step 8: Iteration 3

Sample $X_1^{(3)}$ from $P(X_1 \mid X_2 = 0)$:

$$P(X_1 = 0 \mid X_2 = 0) = \frac{2}{3}.$$

Use $r_7 = 0.61$. Since $0.61 \leq \frac{2}{3}$,

$$X_1^{(3)} = 0.$$

Now sample $X_2^{(3)}$ from $P(X_2 \mid X_1 = 0)$:

$$P(X_2 = 0 \mid X_1 = 0) = \frac{1}{3}.$$

Use $r_8 = 0.42$. Since $0.42 > \frac{1}{3}$,

$$X_2^{(3)} = 1.$$

So

$$(X_1^{(3)}, X_2^{(3)}) = (0, 1).$$

Answer for Part (b)

The Gibbs samples are

$$\boxed{(X_1^{(0)}, X_2^{(0)}) = (1,0), \quad (X_1^{(1)}, X_2^{(1)}) = (0,0), \quad (X_1^{(2)}, X_2^{(2)}) = (1,0), \quad (X_1^{(3)}, X_2^{(3)}) = (0,1).}$$

Why the Procedure Works

Gibbs sampling does not sample from the full joint distribution directly. Instead, it alternates:

- sample X_1 from $P(X_1 \mid X_2)$,
- then sample X_2 from $P(X_2 \mid X_1)$.

These conditional updates define a Markov chain whose limiting distribution is the target joint distribution.

Exam Shortcut

For a small Gibbs-sampling question:

1. Write the joint table clearly.
2. Compute marginals.
3. Compute the conditional tables once.
4. Use the random numbers in order, comparing each one with the relevant cumulative probability.

For binary variables, each update is just a threshold test.

Sources Used

- Assignment 4 PDF, Question 8.
- Local submitted Assignment 4 PDF checked for consistency of the conditionals and sample path.
- Chapter 7 slides on Gibbs sampling and conditional sampling.
- Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 17 on MCMC and repeated conditional updates.